



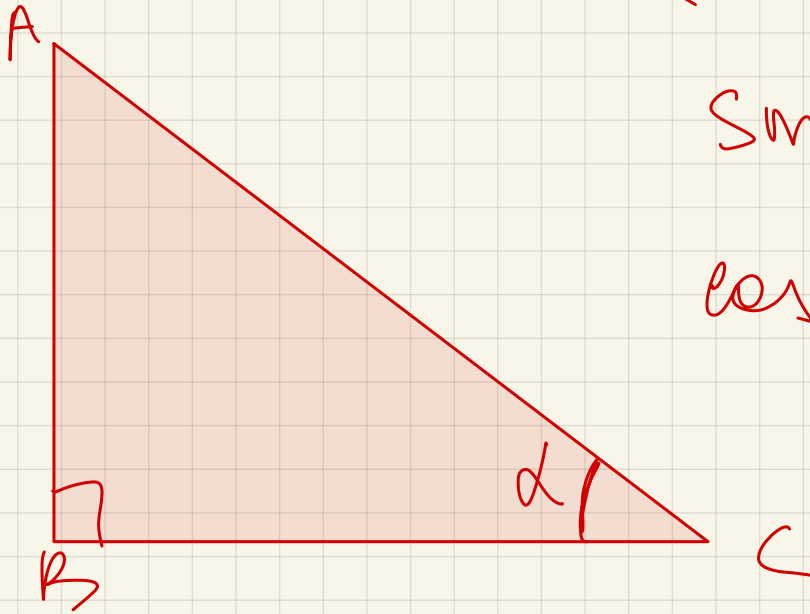
300m

400m

400

$$P = 400 + 900 + 400$$

$$= \underline{\underline{1500m}}$$



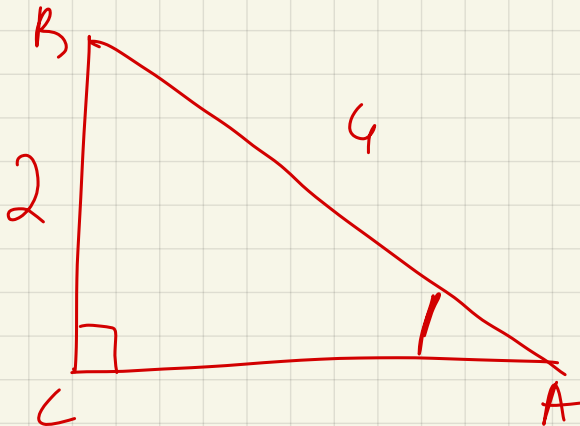
$$\sin \alpha = \frac{AB}{AC}$$

$$\cos \alpha = \frac{BC}{AC}$$

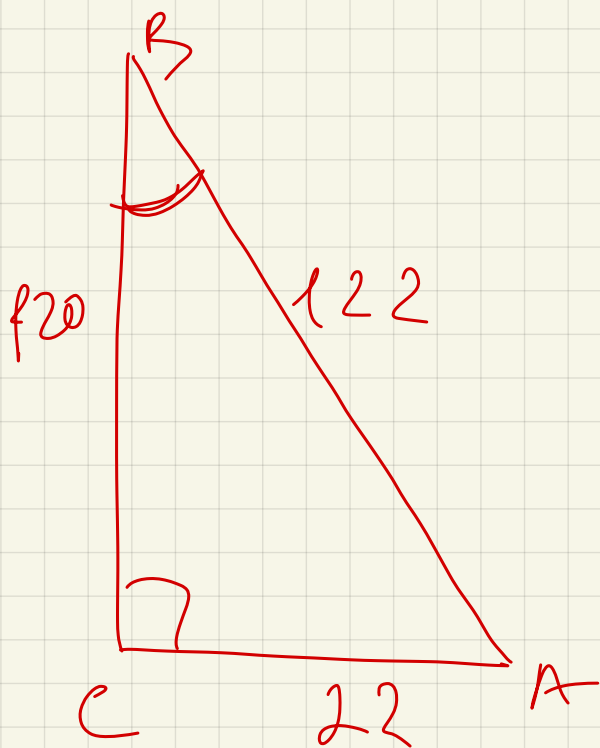
$$\tan \alpha = \frac{AB}{BC}$$

$$\cot \alpha = \frac{BC}{AB}$$

$$AB^2 + BC^2 = AC^2$$



$$\sin A = \frac{2}{4} = \frac{1}{2}$$



$$\cos B = \frac{60}{61}$$

$$AB = 122$$

$$S' = ?$$

$$S = \frac{BC \cdot AC}{2}$$

$$\cos B = \frac{BC}{AB}$$

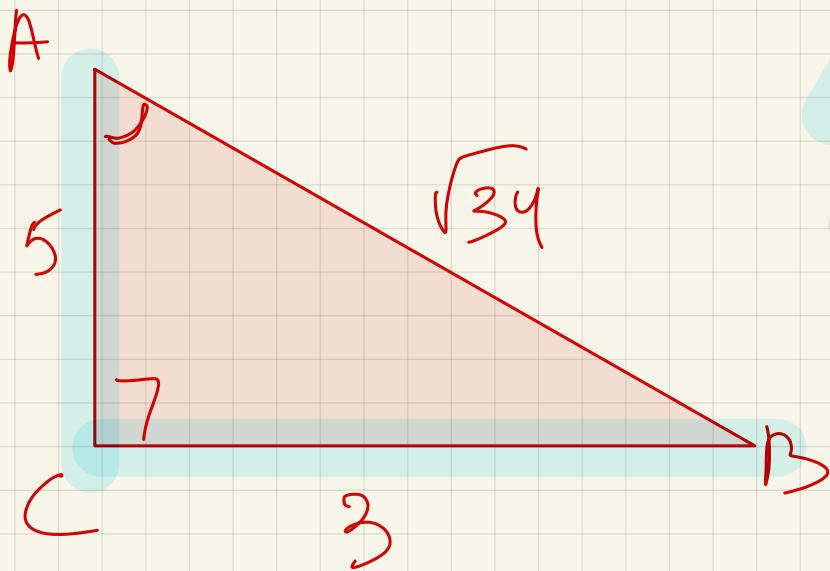
$$\frac{BC}{AB} = \frac{60}{61}$$

$$BC = AB \cdot \frac{60}{61} = 122 \cdot \frac{60}{61} = 120$$

$$BA^2 - BC^2 = AC^2 = 22$$

$$14884 - 14400$$

$$S' = \frac{120 \cdot 22}{2} = 1320$$



$$AB = \sqrt{34}$$

$$BC = 3$$

$$\lg A ?$$

$$S ?$$

$$AC^2 + BC^2 = AB^2$$

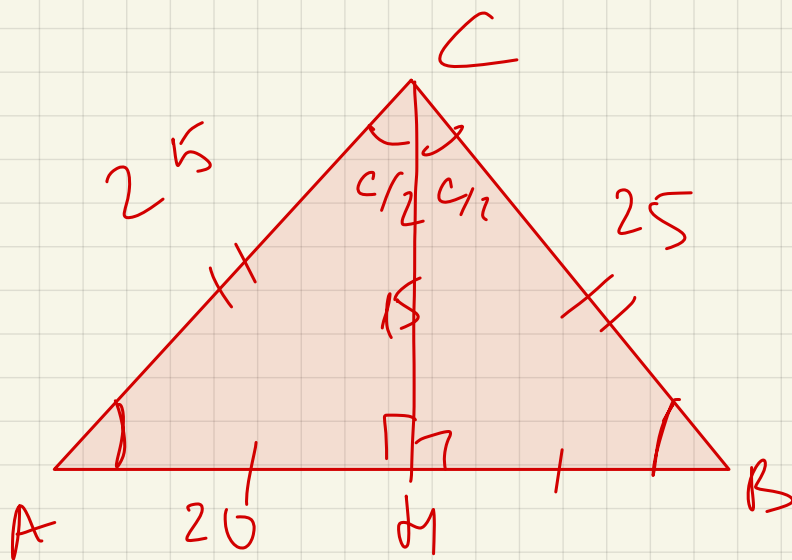
$$AC^2 = 25$$

$$AC = 5$$

$$\lg A = \frac{CB}{AC}$$

$$= \frac{3}{5}$$

$$S = \frac{3 \cdot 5}{2} = 7,5$$



$$\angle A = \angle B$$

CH — высота,
медиана,
биссектриса.

$$AC = 25$$

$$AB = 40$$

$$\sin A ?$$

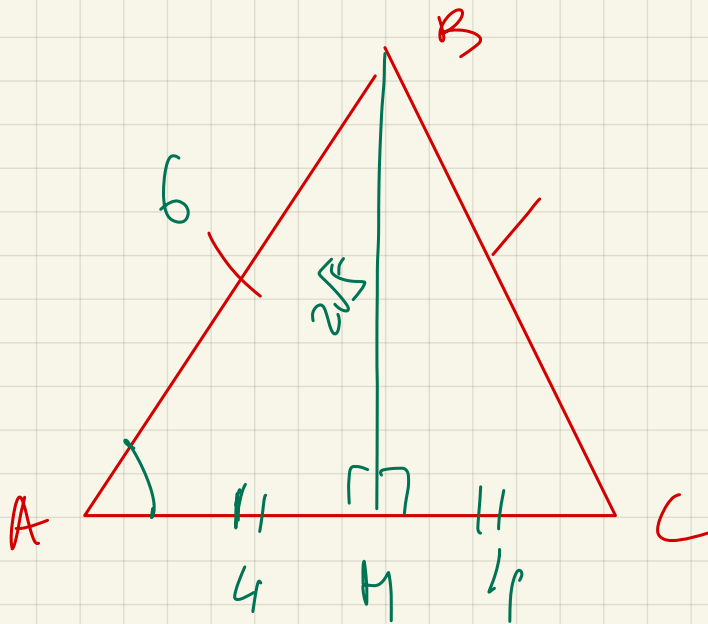
$$AH = \frac{AB}{2} = 20$$

$$AC^2 - AH^2 = CH^2$$

$$= 225 \Rightarrow$$

$$CH = 15$$

$$\sin A = \frac{CH}{AC} = \frac{15}{25} = \frac{3}{5} = \underline{0.6}$$



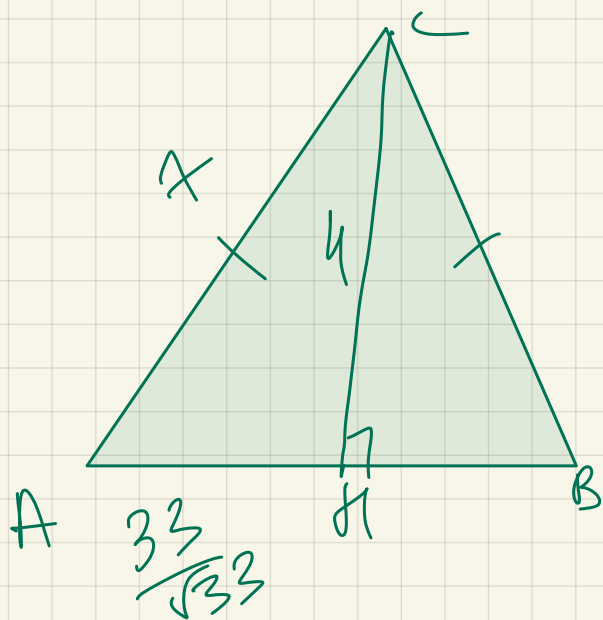
$$\begin{aligned} \tan BAC &= \frac{\sqrt{5}}{2} \\ AC &= 8 \\ AB? \end{aligned}$$

$$\frac{BH}{AN} = \frac{\sqrt{5}}{2}$$

$$BH = AN \cdot \frac{\sqrt{5}}{2} = \frac{4 \cdot \sqrt{5}}{2} = 2\sqrt{5}$$

$$(2\sqrt{5})^2 = 4 \cdot 5 = 20$$

$$AN^2 + BH^2 = AB^2$$



$$CH = 4$$

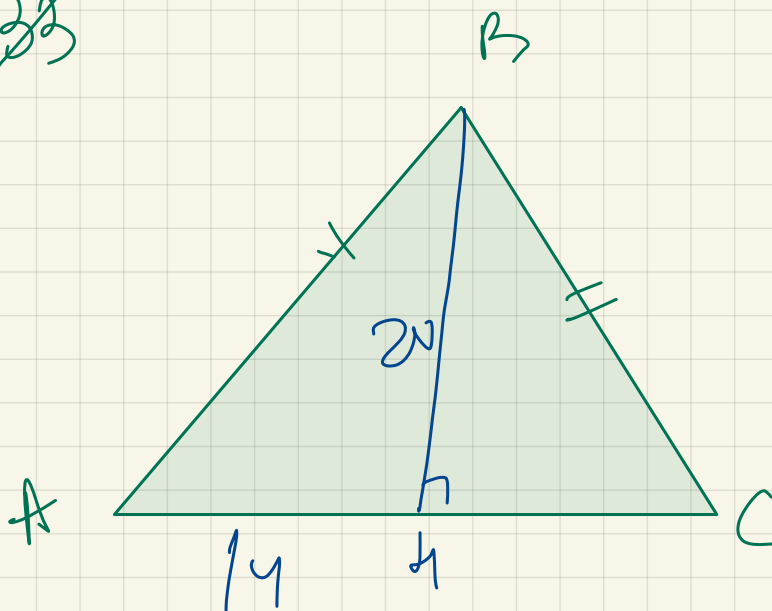
$$\lg A = \frac{4\sqrt{33}}{33}$$

$AC?$

$$\frac{CH}{AH} = \lg A = \frac{4\sqrt{33}}{33}$$

$$\frac{CH}{\lg A} = AH = \frac{33}{\sqrt{33}}$$

$$\frac{33^2}{33}$$



$$\lg A = \frac{10}{7}$$

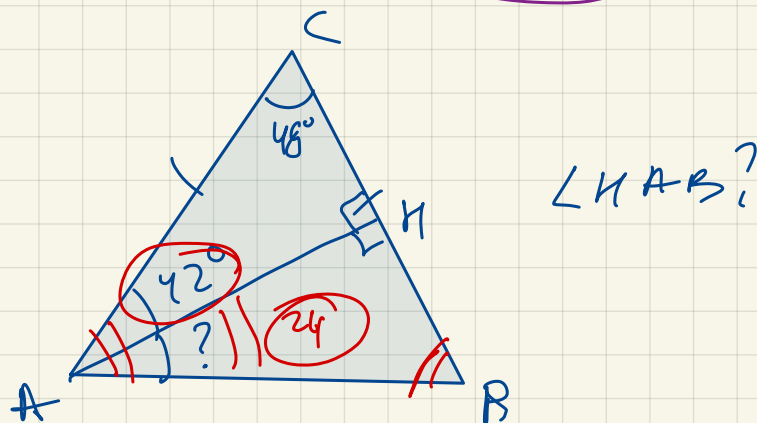
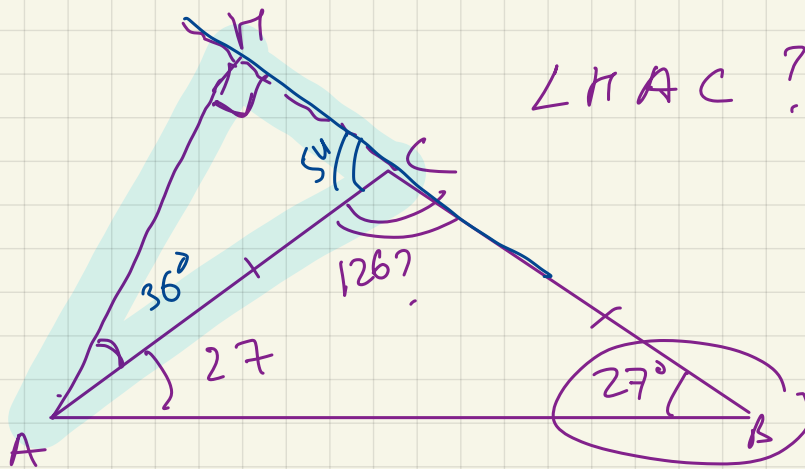
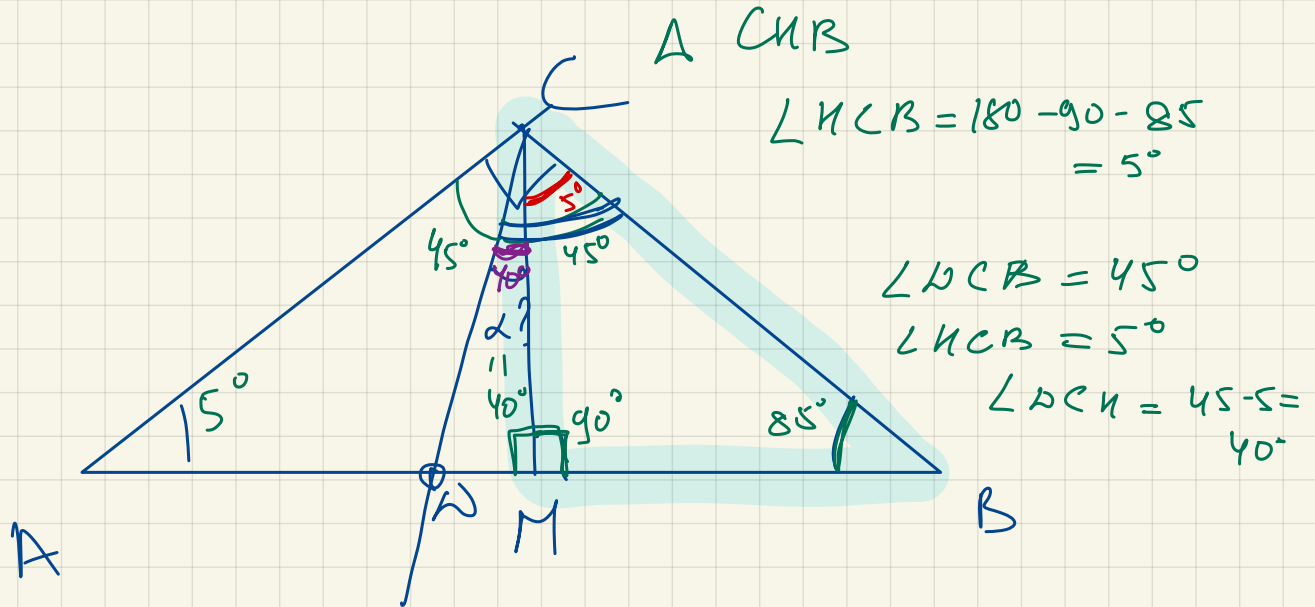
$$AC = 28$$

$$S_{ABC}?$$

$$S = \frac{20 \cdot 28}{2} = \underline{\underline{280}}$$

$$\sum \angle_{\Delta} = 180^{\circ}$$

$$\sum \angle_{\square} = 360^{\circ}$$



$$\angle B + \angle A + 48^\circ = 180$$

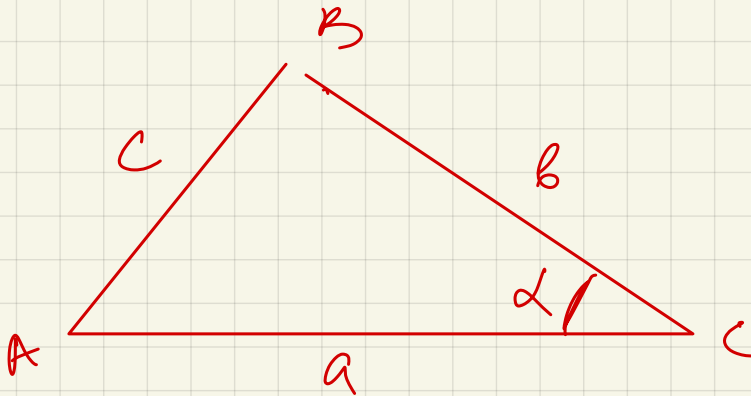
$$2\angle B + 48^\circ = 180^\circ$$

$$2\angle B = 180 - 48 = 132$$

$$\angle B = 66$$

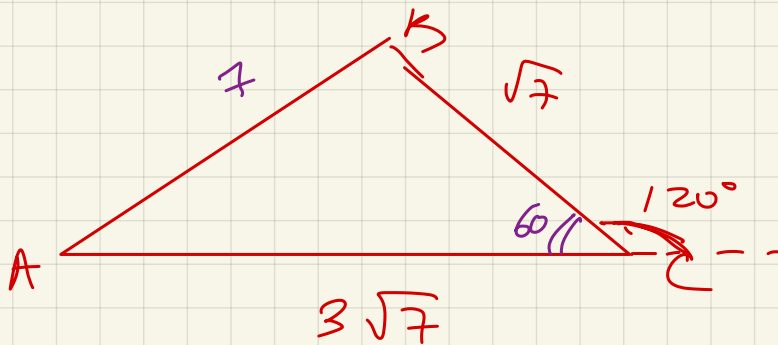
$$\angle A = 66$$

теорема косинусов.



$$c^2 = a^2 + b^2 - 2 \cdot a \cdot b \cdot \cos \alpha$$

$$c = \sqrt{a^2 + b^2 - 2ab \cos \alpha}$$



AB?

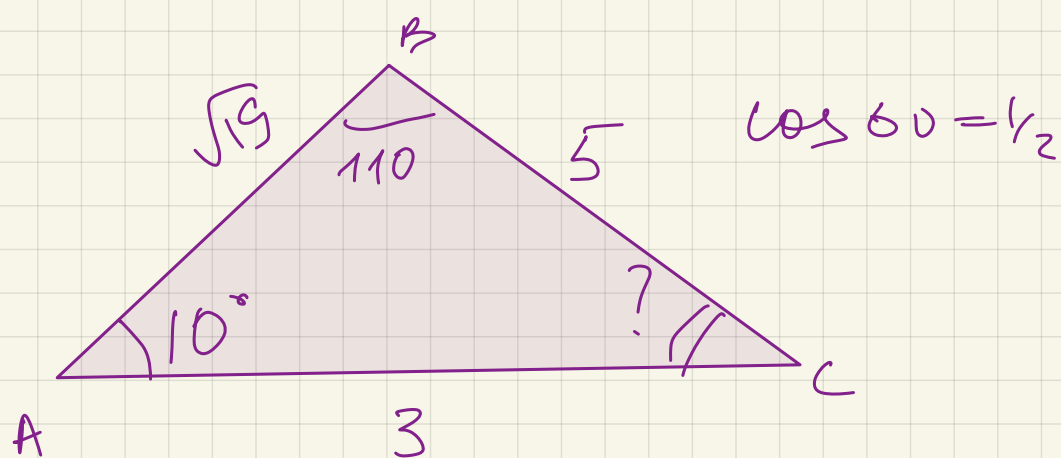
$$\cos 60 = \frac{1}{2}$$

$$AB^2 = BC^2 + AC^2 - 2 \cdot BC \cdot AC \cdot \cos 60^\circ$$

$$7 + 63 - 2 \cdot 3\sqrt{7} \cdot \sqrt{7} \cdot \frac{1}{2}$$

$$70 - 21 = 49$$

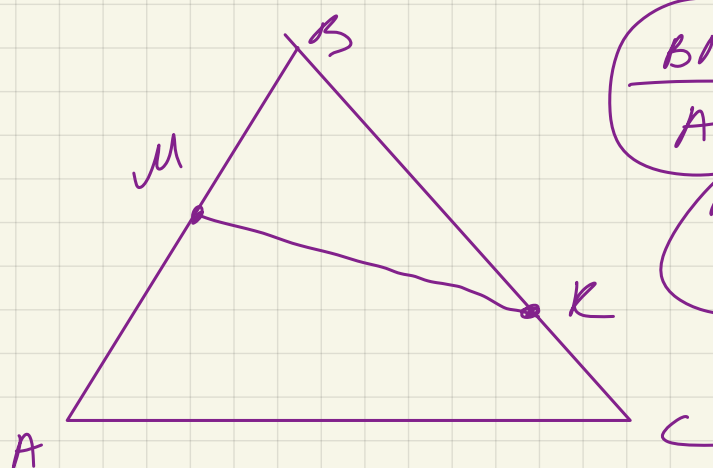
$$AB^2 = 49 \quad AB = 7$$



AB?

$$9 + 25 - 3 \cdot 5 \cdot \frac{1}{2} =$$

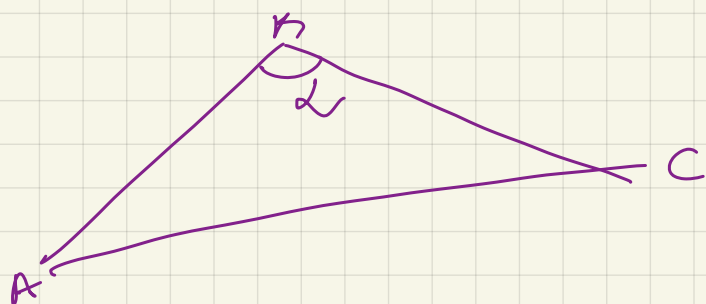
$$9 + 25 - 15 = 19$$



$$\frac{BM}{AB} = \frac{1}{2}$$

$$\frac{BK}{BC} = \frac{4}{5}$$

$$\frac{S_{ABC}}{S_{MBK}} = ?$$



$$S' = \frac{1}{2} AB \cdot BC \cdot \sin B$$

$$S'_{ABC} = \frac{1}{2} AB \cdot BC \cdot \sin B$$

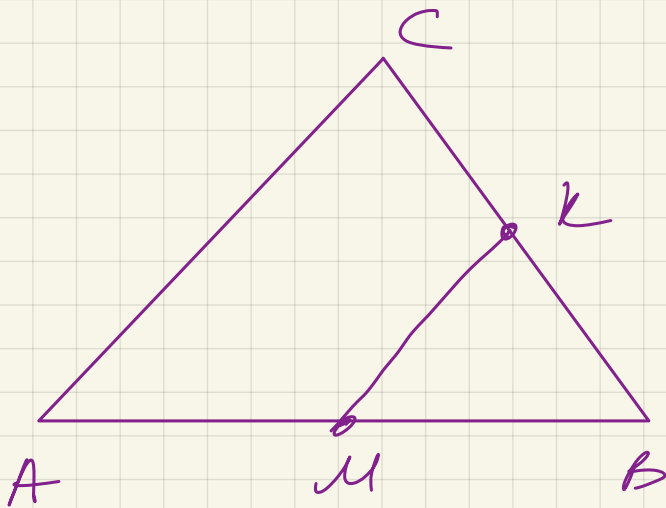
$$S'_{MBK} = \frac{1}{2} MB \cdot BK \cdot \sin B$$

$$\frac{S_{ABC}}{S_{MBK}} = \frac{\frac{1}{2} AB \cdot BC \sin B}{\frac{1}{2} MB \cdot BK \sin B} \left(\frac{AB \cdot BC}{MB \cdot BK} \right) \quad \textcircled{=}$$

$$\frac{AB}{MB} = 2$$

$$\textcircled{=} 2 \cdot \frac{5}{4} = \frac{5}{2} = 2.5$$

$$\frac{BC}{BK} = \frac{5}{4}$$



$$\frac{BK}{BC} = \frac{2}{3}$$

$$\frac{BM}{AB} = \frac{1}{2}$$

$$\frac{S_{ABC}}{S_{MBK}} = ?$$

$$\frac{AB \cdot BC}{BK \cdot BM} = 2 \cdot \frac{3}{2} = 3$$